8. Which of the following could be the nth partial sum for the infinite series $\sum_{n=1}^{\infty} \frac{1}{4^n}$?

Σ

$a = 1$

$r = \frac{1}{4}$

(A) $S_n = \frac{1}{3}\left(1 + \frac{1}{4^n}\right)$ (B) $S_n = \frac{1}{3}\left(1 - \frac{1}{4^{n+1}}\right)$ (C) $S_n = \frac{1}{3}\left(1 - \frac{1}{4^n}\right)$ (D) $S_n = \frac{1}{4}\left(1 - \frac{1}{3^n}\right)$

$$S_n = \frac{1}{4^1} + \cancel{\frac{1}{4^2}} + \cancel{\frac{1}{4^3}} + \cancel{\frac{1}{4^4}} + \cancel{\cdots} + \cancel{\frac{1}{4^n}}$$

$$\frac{1}{4} S_n = \cancel{\frac{1}{4^2}} + \cancel{\frac{1}{4^3}} + \cancel{\frac{1}{4^4}} + \cancel{\frac{1}{4^5}} + \cancel{\cdots} + \cancel{\frac{1}{4^n}} + \frac{1}{4^{n+1}}$$

$$S_n - \frac{1}{4} S_n = \frac{1}{4} - \frac{1}{4^{n+1}}$$

$$\frac{S_n \cancel{\left(1 - \frac{1}{4}\right)}}{\cancel{\left(1 - \frac{1}{4}\right)}} = \frac{\frac{1}{4} - \frac{1}{4^{n+1}}}{\left(1 - \frac{1}{4}\right)}$$

$$S_n = \frac{\frac{1}{4} - \frac{1}{4^{n+1}}}{\frac{3}{4}}$$

$$S_n = \left[\frac{1}{4} - \frac{1}{4^{n+1}}\right] \frac{4}{3}$$

$$S_n = \frac{\cancel{4}^{1}}{\cancel{12}_{3}} - \frac{4}{3(4^{n+1})}$$

$$S_n = \frac{1}{3} - \frac{1}{3}\left(\frac{4}{4^{n+1}}\right)$$

$$S_n = \frac{1}{3}\left[1 - \frac{4}{4^{n+1}}\right]$$

$$\downarrow$$

$$\frac{\cancel{4}}{4^n \cdot \cancel{4}}$$

$$\frac{1}{4^n}$$

$$\boxed{S_n = \frac{1}{3}\left[1 - \frac{1}{4^n}\right]}$$